AP CALCULUS AB

Unit 1 • Worksheet 1-5A

# Worksheet 1-5A • IVT and EVT

Name: _____ Date: _____

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Question 1

NO CALCULATOR

$f$ is continuous on $[-1, 3]$, with $f(-1) = 5$ and $f(3) = -2$. Which must be true?

(A) $f(1) = 1.5$

(B) $f(c) = 0$ for some $c$ in $(-1, 3)$

(C) The minimum of $f$ is $-2$.

(D) $f$ is decreasing.

## Question 2

NO CALCULATOR

$f$ is continuous on $[2, 5]$, with $f(2) = 3$ and $f(5) = 11$. The IVT guarantees that $f(c)$ equals which value for some $c$ in $(2, 5)$?

(A) 12

(B) 2

(C) 7

(D) 0

## Question 3

NO CALCULATOR

$h$ is continuous.

| $x$ | 0 | 1 | 3 | 4 | 6 |
|---|---|---|---|---|---|
| $h(x)$ | $-2$ | 3 | 1 | $-1$ | 2 |

What is the least number of zeros of $h$ on $[0, 6]$?

(A) 1

(B) 2

(C) 3

(D) 4

## Question 4

NO CALCULATOR

Which function is guaranteed to have an absolute maximum on the given interval?

(A) $x^3$ on $[0, 2]$

(B) $\frac{1}{x}$ on $(0, 1]$

(C) $x^2$ on $(-1, 1)$

(D) $\frac{1}{x-3}$ on $[2, 4]$

### Question 5

NO CALCULATOR

Which interval matches $1 \leq x < 4$?

(A) $(1, 4)$
(B) $[1, 4]$
(C) $(1, 4]$
(D) $[1, 4)$

### Question 6

NO CALCULATOR

What does $f(c) \leq f(x)$ for every $x$ in $[a, b]$ mean?

(A) $f$ has an absolute minimum at $x = c$.
(B) $f$ has an absolute maximum at $x = c$.
(C) $f(c) = 0$.
(D) $f$ is decreasing.

### Question 7

NO CALCULATOR

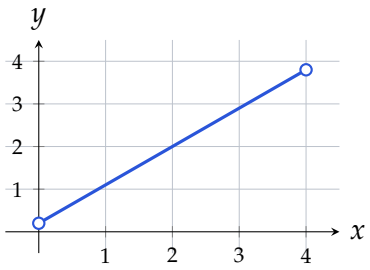
$f(x) = \frac{1}{x}$ has $f(-1) = -1$ and $f(1) = 1$, but $f(x)$ is never $0$. Why doesn't this contradict the IVT?

(A) $0$ is not between $-1$ and $1$.
(B) The interval is open.
(C) $f$ is not continuous on $[-1, 1]$.
(D) The IVT only works for polynomials.

### Question 8

NO CALCULATOR

The graph of $f$ on the open interval $(0, 4)$ is shown. Which is true?




(A) $f$ has a maximum at $x = 4$.
(B) $f$ has a maximum and a minimum on $(0, 4)$.
(C) $f$ has no maximum and no minimum on $(0, 4)$.
(D) The EVT guarantees a maximum.

**Question 9** (Free response) NO CALCULATOR

The temperature $T(t)$, in °C, is continuous.

| $t$ (h) | 0 | 2 | 5 | 8 |
|---|---|---|---|---|
| $T(t)$ | 12 | 18 | 15 | 9 |

(a) Must $T(t) = 16$ for some $t$ in $(0, 2)$? Justify.
(b) What is the least number of times $T(t) = 16$ on $[0, 8]$? Explain.
(c) Must $T$ have an absolute maximum on $[0, 8]$? What can you say about its value?